MaCVi 2026 — Embedded Semantic Segmentation Challenge

Preliminary Strategy Report — February 21, 2026

This report documents initial strategic analysis for the MaCVi 2026 Embedded Semantic Segmentation Challenge, based on research into the challenge rules, the MaCVi 2025 results paper, and the LaRS dataset. It covers key questions around dataset strategy, model selection, and quantization, along with additional recommendations. This is a living document that will be expanded as the approach evolves.

Context

The MaCVi Embedded Semantic Segmentation Challenge requires building a model that segments maritime scenes into three classes (sky, water, obstacle), exported as ONNX, automatically quantized to INT8, and deployed on a Luxonis RVC4 embedded device. Models must run at ≥30 FPS with 768×384 input. The winner is determined by the F1 detection metric, with FPS used as tiebreaker. The primary training dataset is LaRS (2,607 train / 200 val / 1,205 test images), though external public datasets are permitted with disclosure. The mmsegmentation-macvi repository is provided as a starter codebase.

An initial strategy discussion was held to address several key questions before beginning experimentation. The central ideas explored were:

1. Whether investing in dataset engineering (combining multiple public maritime datasets) would yield better results than focusing on model architecture.
2. How to handle label format mismatches when merging datasets.
3. Whether using existing model architectures is permitted or if novel architectures are expected.
4. Whether Quantization-Aware Training is worth the added complexity given the automatic INT8 quantization pipeline.

There was also a prior attempt at this challenge for MaCVi 2025 that started too late to iterate effectively, so a key goal this time is to plan early and use tools like Claude Code agents to accelerate time-consuming engineering tasks (dataset scripts, training configs, etc.). The following sections capture the analysis and recommendations for each of these questions.

1. On the "Data Matters More Than the Model" Hypothesis

> ***Question to consider: Is it true that a better dataset with a simpler model can outperform a bigger SOTA model with a worse or smaller dataset? Is this a valid foundation for the challenge strategy?***

This intuition has real merit, but it needs nuance. The idea that data quality and diversity matter more than raw model complexity is well-supported in practice—especially in domain-specific tasks like maritime segmentation where the distribution is narrow and specialized. A well-curated, representative training set with clean labels will almost always beat throwing a massive model at noisy or insufficient data.

However, there is an important caveat for this specific challenge: **the evaluation is on the LaRS test set only.** This means the question is not just "does more data help in general?" but "does more data help the model generalize better *on LaRS's distribution specifically*?" These are different questions. Adding external

data can help if it fills gaps in LaRS's training distribution (e.g., rare lighting conditions, unusual obstacle types, underrepresented scenes). But it can also **hurt** if the external data introduces distribution shift—different camera perspectives, different water appearances, different annotation standards—that pulls the model's learned features away from what LaRS expects.

Looking at the MaCVi 2025 results: of the top 3 methods in obstacle segmentation, two (WaterFormer, 1st place, and Advanced K-Net, 3rd place) used LaRS data only. The one that did use external data (WSFormer, 2nd place) used WaterScenes for **pre-training**, not mixed training—then fine-tuned on LaRS. So of the top 3, two used LaRS only, and the one that did use external data used it for pre-training, not mixed training. That is a telling signal.

Recommendation: Data quality matters, but for a LaRS-only evaluation, model architecture choice, training strategy, augmentation, and training at target resolution are likely higher-ROI than simply adding more data. External data should be treated as a secondary optimization, not the primary strategy.

2. External Dataset Integration Methodology

> ***Question to consider: Since the challenge allows external datasets with disclosure, and other datasets will not have the exact same 3-class masks as LaRS, what is the best approach to integrate them? Two options were identified: (1) remapping existing masks into the LaRS format, or (2) using a large zero-shot segmentation model to generate masks. A LaRS-only baseline would be established first. Is this methodology sound?***

Option 1 (mask remapping) is the cleaner path. WaterScenes has semantic segmentation annotations with categories like pier, buoy, sailor, ship, boat, vessel, kayak, and free-space. All of those can be remapped: object categories → obstacle, water-related → water, and sky → sky. WaterScenes has ~54,000 images, which is a massive increase. The key risk is that WaterScenes was captured primarily in Chinese inland waterways, so the visual distribution (water color, typical obstacles, scene composition) may differ significantly from LaRS's globally diverse maritime scenes. Also worth considering is **MASTR1325**, which already has the exact 3-class annotation (sea, sky, environment/obstacle) and is specifically from coastal USV perspectives—very close to LaRS's domain.

Option 2 (zero-shot segmentation model to generate masks) is riskier. Training on pseudo-labels introduces noise. A large model like SAM or GroundedSAM could generate reasonable masks, but there would inevitably be systematic errors that the model would learn from. This should only be pursued if Option 1 is not feasible for a given dataset.

Critical recommendation for both options: Do not just mix everything into one training set. Use a staged approach—pre-train on the larger/external data, then fine-tune exclusively on LaRS. This lets the model learn general water/sky/obstacle features first, then adapt to LaRS's specific distribution and annotation style.

The baseline-first approach is absolutely the right call. Without a LaRS-only reference point, there is no way to objectively measure whether adding more datasets is actually improving the model.

3. Will External Data Actually Help Given LaRS-Only Evaluation?

> ***Question to consider: Since the challenge evaluates the model exclusively against LaRS, and LaRS already has 2,607 training images for a 3-class task, would adding datasets actually***

provide any significant benefit?

With 2,607 training images, LaRS is not in severe data-scarcity territory for a 3-class segmentation task—that is actually a reasonable amount. The marginal benefit of extra data depends on:

- **Are there failure modes in the LaRS-only model?** Look at per-attribute performance (reflections, glitter, night scenes, small obstacles). If the model struggles on specific conditions that external data could address, it is worth it. If it is already strong across the board, more data may just add noise.
- **How different is the external data?** The more similar to LaRS, the more it helps. MASTR1325 > WaterScenes > random maritime imagery, in terms of domain alignment.
- **Diminishing returns vs. engineering time.** Given time constraints, the hours spent writing mask conversion scripts and debugging data pipelines might be better spent on model architecture choices, augmentation strategies, and training at the target resolution (768×384).

Practical advice: Start with LaRS only, get a strong baseline, then try pre-training on MASTR1325 + WaterScenes → fine-tune on LaRS as a single experiment. If it helps, great. If not, not much time is lost.

4. Model Selection: Existing Architectures Are Fine

Question to consider: Does the challenge require creating novel model architectures from scratch, or is it acceptable to grab existing models and train them with a refined methodology?

Using existing model architectures is absolutely allowed. The challenge page explicitly points to the mmsegmentation-macvi repository as a starting point, which contains implementations of various established segmentation models. Looking at the MaCVi 2025 embedded track winners:

- **RSOS-Net:** ResNet-101 backbone + lightweight decoder with separable convolutions and attention-based fusion. Custom but built from standard components.
- **EFFNet:** Ensemble of ResNet-50 and ResNet-101 with DeepLabv3+ decoders. Entirely standard architectures, no novelty in model design.
- **eWaSR-RN50/101:** Previous year's winner, also based on standard ResNet backbones.

Nobody in the embedded track invented a novel architecture from scratch. This is a challenge about engineering the best solution—selecting proven architectures and optimizing them for the constraints (FPS threshold, INT8 quantization robustness, target resolution). Pick architectures from the mmsegmentation-macvi repo, train them well, and focus innovation on training strategy, augmentation, and quantization-robustness.

5. INT8 Quantization and Quantization-Aware Training (QAT)

Question to consider: Since submitted models are quantized to INT8 for evaluation on the embedded device, would Quantization-Aware Training (QAT) be worth the added complexity?

The MaCVi 2025 challenge organizers noted that some models experienced "substantial performance drops" after quantization and that selecting encoders that maintain stability post-quantization is a promising direction. The official guidance says to use FP32 during training and that most performance drop comes from lower input resolution rather than quantization itself.

However, this does not mean QAT is pointless—it means the organizers have not seen many teams try it yet, which is an opportunity. Here is the breakdown:

- **Post-Training Quantization (PTQ)** is what most participants default to (train in FP32, export ONNX, let the platform quantize to INT8). This works fine for quantization-friendly architectures (ResNets, MobileNets).
- **QAT could provide an edge** specifically for architectures that are borderline—models with operations or activation distributions that do not quantize cleanly. If a more powerful architecture is chosen that is close to the FPS threshold, QAT might let it squeeze out better accuracy while still surviving INT8 quantization.
- **The practical challenge** is that QAT adds complexity to the training pipeline, and the quantization is done by Qualcomm's SNPE toolkit on the Luxonis RVC4 device, not by custom code. So the quantization during training would need to be simulated in a way that aligns with how SNPE actually quantizes. This might not be trivial.

Recommendation: Start without QAT. Get the best PTQ result first. If a significant accuracy drop is observed after quantization on the validation set (can be simulated locally), then invest in QAT. Do not over-engineer before there is data showing it is needed.

6. Additional Recommendations

1. Train at target resolution (768×384). This was called out explicitly by the organizers and by multiple winning teams. The resolution mismatch between training and deployment is the biggest source of accuracy drop, even more than quantization.

2. Focus on the F1 detection metric, not just mIoU. The winner is determined by F1 (obstacle detection quality), with FPS used as tiebreaker. This means correctly detecting obstacles matters more than pixel-perfect boundaries. Think about what this implies for loss function and augmentation strategy.

3. Augmentation is likely the highest-ROI lever. RandomResize, RandomCrop, RandomFlip, PhotoMetricDistortion—these were used by virtually all top teams. Maritime-specific augmentations like simulating sun glitter, reflections, and varying water color could help too.

4. Consider the 4-class trick. WaterFormer, the top method in MaCVi 2025 obstacle segmentation, trained with 4 classes by splitting obstacles into sub-categories, then merged them back for evaluation. This forces the model to learn more discriminative features within the obstacle class. Worth experimenting with.

5. Test Time Augmentation (TTA)—several top teams used this. For the embedded track, TTA is likely too expensive (the 30 FPS requirement), so skip it, but keep it in mind for the regular segmentation track if a submission is also made there.

6. Claude Code agents—good candidates for delegation include: dataset conversion/remapping scripts, config file generation, ONNX export scripts, experiment tracking boilerplate, augmentation pipeline code, and evaluation/visualization utilities. This frees focus for the experimental decisions rather than mechanical implementation work.

7. Backbone selection matters for quantization. ResNet-based backbones tend to quantize very cleanly to INT8. More exotic architectures (Vision Transformers, DINOv2) may have more issues. For the embedded track specifically, lean toward ResNet-50/101 or similar CNN-based backbones with

DeepLabv3+ or similar decoders, rather than trying to squeeze a transformer onto the device.

Overall, the plan is solid. The main priorities in order are: (1) a strong LaRS-only baseline trained at 768×384, (2) smart augmentation, (3) architecture selection optimized for INT8 quantization, and (4) optionally pre-training on external data and fine-tuning on LaRS. Good luck with MaCVi 2026!

End of Preliminary Report